

**M.TECH
(SEM I) THEORY EXAMINATION 2022-23
RESEARCH PROCESS & METHODOLOGY**

Time: 3 Hours**Total Marks: 70****Note:** Attempt all Sections. If require any missing data; then choose suitably.**SECTION A****1. Attempt all questions in brief.****2 x 7 = 14**

- (a) Describe the classification of research design.
- (b) Explain null and alternative hypothesis.
- (c) Discuss the steps of research process
- (d) Distinguish probability and non-probability methods of sampling techniques.
- (e) Explain research method and research methodology.
- (f) Discuss the objectives of a research.
- (g) Evaluate research report.

SECTION B**2. Attempt any three of the following:****7 x 3 = 21**

- (a) Describe research problem and evaluate the techniques used to define a research problem.
- (b) Explain the meaning and significance of research design. Evaluate Random Replication and Randomized Block research designs.
- (c) Discuss stratified random sampling design with a suitable example. How to select such a sample?
- (d) Describe a hypothesis and discuss the parametric tests in detail.
- (e) Explain the following:
 - (i) Indexing and citation of journals
 - (ii) Intellectual property

SECTION C**3. Attempt any one part of the following:****7 x 1 = 7**

- (a) Evaluate research and its significance.
- (b) Write short notes on:
 - (i) Indexing & citation of journals.
 - (ii) Plagiarism
 - (iii) Referencing Styles
 - (iv) SPSS

4. Attempt any one part of the following:**7 x 1 = 7**

- (a) Explain hypothesis testing with an example.
- (b) Enumerate the methods of data collection. Differentiate between collection of data through questionnaires and schedules.

5. Attempt any one part of the following:**7 x 1 = 7**

- (a) Discuss the before-and-after with and without control design in context of a research design.
- (b) Distinguish case study method and the survey method. Explain the merits and

limitations of case study method.

6. Attempt any *one* part of the following:

7 x 1 = 7

- (a) Research design in exploratory studies must be flexible but in descriptive studies, it must minimize bias and maximize reliability”. Discuss.
- (b) Discuss the Karl Pearson’s coefficient of correlation. Calculate the Karl Pearson’s coefficient of correlation from the following data:

Roll No. of students	Marks in Math	Marks in Science
1	48	45
2	35	20
3	17	40
4	23	25
5	47	45

7. Attempt any *one* part of the following:

7 x 1 = 7

- (a) Explain the significance of research report and narrate the various steps involved in writing such a report.
- (b) What is a chi-square test? Genetic theory states that children having one parent of blood type A and the other of blood type B will always be one of three types, A, AB, B and that the proportion of three types will on an average be as 1:2:1. A report states that out of 300 children having one A parent and B parent, 30% were found to be types A, 45 % type AB and remainder type B. Test the hypothesis by chi- square test. (Given: chi-square for 2 d.f. at 5%level of significance= 5.991)